

Nanotechnology joining hands with the IoT

Sensors and actuator units make an IoT device a functional system that is capable of monitoring correct physical parameters and responding correspondingly. These sensors and actuators are prevalent at macro-scale in industrial applications, and these technologies cannot be scaled for numbers that IoT business-roadmaps foresee. Folks at Centre for Nano Science and Engineering (CeNSE) at Indian Institute of Science (IISc), Bengaluru, assure that nanotechnology will help build novel sensors and actuators in IoT domains, such as ambient monitoring, automobile, health monitoring of buildings, biomedical applications and many more.

Bulk sensors are replaced with components that are sensitive, rugged, stable, scalable, economical, have small form-factor and low-power, making it viable for the IoT. "Furthermore, nanotechnology promises to integrate sensors, actuators, processors, radios and energy-harvesting devices to make the Internet of nano-enabled Things in the near future for more potential applications," say Dr Vijay Mishra and A. Peter Manoj, both from CeNSE.

This means that the system will now have to only transmit results to the central system, resulting in a lower amount of transmitted data. Conventional processing solutions are unable to crunch Big Data fast enough to keep-up with the never-ending flow of incoming data.

Bryan Fletcher, technical marketing director, and Ramani Sundesan, India managing director, both at Avnet, mentioned how programmable SoCs could be used to crunch this data by leveraging the massive parallel processing power of field programmable gate arrays (FPGAs) with embedded microprocessors. These complete systems on a programmable chip form a sort of reprogrammable CPU architecture. Some examples of vendors with these kinds of chips are Xilinx Zynq, Altera Arria and ActelSmartFusion families.

Predictive engine diagnostics

Big-time engine makers, like Rolls Royce, BMW and Mercedes Benz, were into Big Data and the IoT business model back in mid 2000s, even before they were a buzzword. Their engine health-monitoring unit combines latest sensor technologies with data collection, management and analyses techniques, letting them accurately predict engine failures at an early stage. This optimises engine maintenance and repair schedule, thereby improving

safety and providing better customer services at lower costs.

Taking one step further in Big Data business model, Rolls Royce has partnered with seven other firms (University of Nottingham, Fraunhofer IPA, IK4 Tekniker, ETH Zurich, AREVA NDE-Solutions, Acciona Infrastructure and OMV Petrom) to develop a snake robot that is equipped with self-positioning, reasoning, planning and adaptation capabilities. The 1.25cm (half-inch) diameter robot named MiRoR (miniaturised robotic systems for holistic in-situ repair and maintenance works in restrained and hazardous environments) can take pictures of engines' interiors and send it in real-time to experts who control it remotely. This will let engine experts quickly find faults in large, complex machines like aircrafts.

Mercedes-Benz is an example of an automotive manufacturer who has embraced the IoT. The roadside assistance, safety and security features provided by this popular car manufacturer have been enhanced with the introduction of mbrace feature in their cars. Their new system now enables remote vehicle controls, performs remote engine diagnostics and delivers software updates to keep your car running perfectly. Of course, Indian electric car manufacturer Mahindra Reva Electric Vehicles has also had some of these features in their e2o car since 2013.

Microlise is another example of firms that provide vehicle and machine telematics solutions. A winner of JCB Supplier Award for Innovation, they monitor fleet performance and understand journey management to deliver a variety of solutions that help end-users reduce fuel consumption and fleet size while maintaining performance. Microlise has also partnered with VEDAT project, which received an enormous amount of funding earlier this year from the UK's innovation agency Technology Strategy Board.

Energy management with Big Data

Several terabytes of data coming from sensors and energy metres can be used for intelligent monitoring of power-usage and increasing the efficiency of the whole system. Smart buildings, a concept of making buildings smarter and energy-efficient using the IoT and Big Data analyses, is already being implemented by Nest Labs.

Their first product is a smart, Wi-Fi-enabled, sensor-driven, programmable, self-adapting thermostat. It monitors the user's temperature adjustments and uses sensors (temperature, humidity and activity sensors) and sophisticated algorithms to learn and identify patterns, which are later used to intelligently control the heating of the home, intuitively. It adjusts the temperature according to the time of the day, weather condition and human activities inside the home, thereby providing minimal and an effective use of energy. Being Wi-Fi-connected, the device follows current weather forecasts and adjusts the room temperature accordingly. It can also be controlled from remote locations using laptops, tablets and smartphones.

Intelligent solar and wind analytics is another area that has seen many advances. Mumbai based two-year-old startup, Algo Engines, uses Big Data analytics to provide operational intelligence for solar power plants, wind turbines and other IoT equipment. Data from various sensor components